

3. **Reserve:** These nodes operate as active nodes, i.e. meets the staking requirements (see 5.1.1) and a machine up and running which also participates in the replication process. When the size of the active set falls below 15, these nodes become active. Initially the target is 6 reserve nodes, such that active and reserve nodes make up 21 nodes.
4. **Standby:** A Node is in Standby status when it meets the staking requirements (see 5.1.1), but the Node Operator is not running a machine. They need to be ready, however, to run a machine when the size of the active and reserve set falls below 21 - if they fail to do so, when called to join the active set, they remain in the standby set.⁴
5. **Deactivate:** A node operator is free to decide to stop supporting the Network by triggering the deactivation event. Nodes in this deactivated set will not receive any more rewards, and need to wait until their timelock decreases to 0 to withdraw their stake.

This mechanism permits only reliable nodes to participate in processing user requests, maintaining the Network’s integrity and performance.

5.1.3 Service Node Performance

Node performance is a critical factor that directly impacts the overall efficiency, reliability, and scalability of the system. To evaluate individual node performance utilizing key metrics to generate a performance coefficient $U(i, t) \in [0, 1]$ for each node i at time t .

The evaluation framework incorporates two primary performance metrics:

1. **Uptime** (U_i): Defined as the percentage of time a node remains operational, calculated as the ratio of total operational time to the evaluation period.
2. **Latency** (L_i): Measured as the average response time for a node to process requests, computed by dividing the sum of response times by the number of messages.

To provide a nuanced assessment, a weighted scoring function is introduced that combines these metrics:

$$\text{Performance} = (W_u \cdot U_i) \cdot (W_l \cdot L_i)$$

Where W_u and W_l represent the weights assigned to uptime and latency, respectively. These weights can be adjusted based on Network priorities and requirements.

The WalletConnect Network employs a dynamic two-phase system to measure and verify node performance, facilitating the transition from a permissioned to a permissionless network.

⁴If in future developments delegation of stake is enabled those that have insufficient delegation would be included in the standby set.